# Programming Fundamentals

## Lab 4

---

**Name:** Muhammad Owais

**Roll Number:** 2

**Program:** BS Computer Science (BSCS)

**Section:** D

**University:** Ghazi University

## Task 1:

Write a program to print positive number entered by the user.

If the user enters negative number print number entered is positive otherwise print number is negative.

### Input

Task_1.cpp

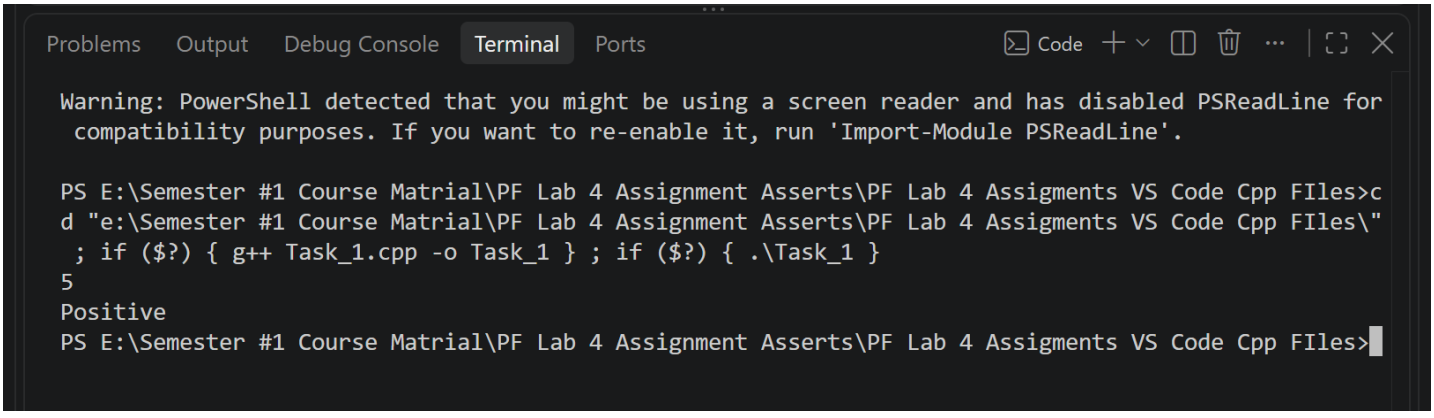
```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

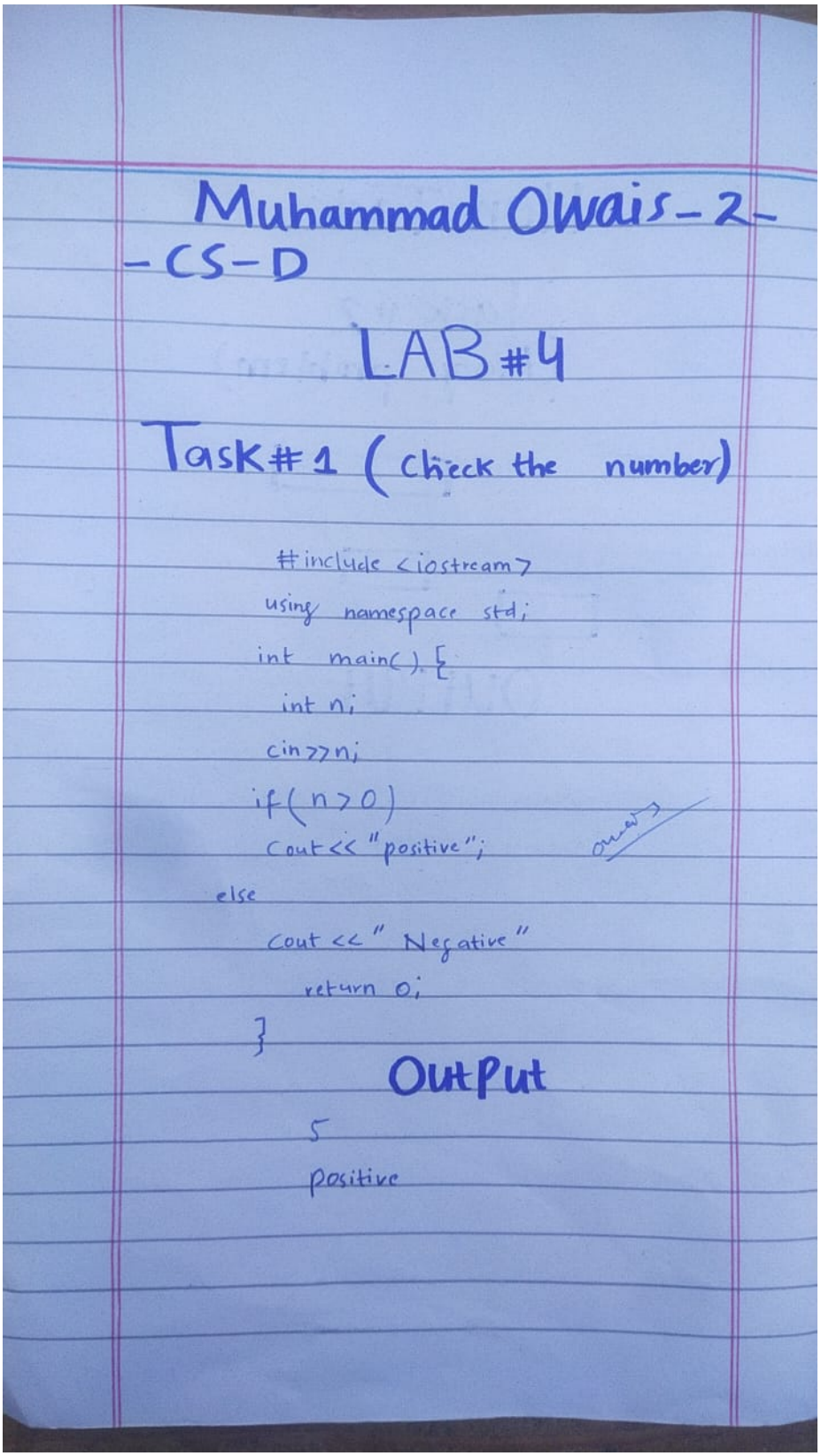
    if (n > 0)
        cout << "Positive";
    else
        cout << "Negative or Zero";

    return 0;
}
```

### Output



**Handwritten Notes**



## Task 2:

Program to check whether an integer is positive, negative or zero.

### Input

Task_1.cpp | Task_2.cpp

Task_2.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;

    if (n > 0)
        cout << "Positive";
    else if (n < 0)
        cout << "Negative";
    else
        cout << "Zero";

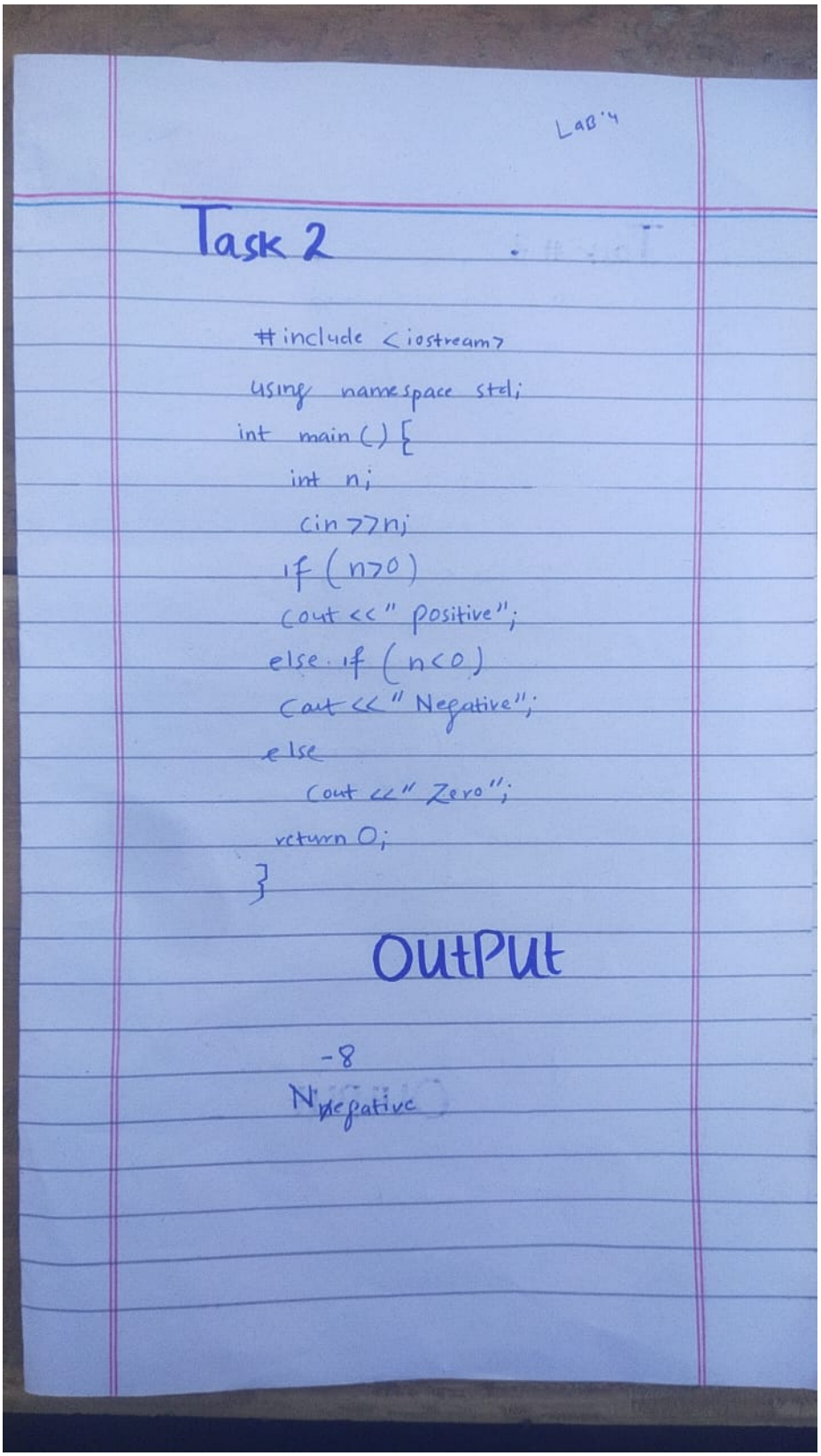
    return 0;
}
```

### Output

Problems | Output | Debug Console | Terminal | Ports

```
PS E:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles>c
d "e:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles\"
 ; if ($?) { g++ Task_2.cpp -o Task_2 } ; if ($?) { .\Task_2 }
-8
Negative
PS E:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles>
```

**Handwritten Notes**



Lab 4

Task 2

```
#include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    if (n > 0)
    cout << " positive";
    else if (n < 0)
    cout << " Negative";
    else
    cout << " Zero";
    return 0;
}
```

OutPut

-8
Negative

## Task 3:

Input : Mark

Process : If mark greater than and equal to 75, score will be A
If mark less than 75 and greater than and equal to 60, score will be B
If mark less than 60 and greater than and equal to 45, score will be C
If mark less than 30, score will be D

Output : Print the grade of your score.

### Input

Task_1.cpp | Task_2.cpp | Task_3.cpp

Task_3.cpp > main()

```cpp
#include <iostream>
using namespace std;

int main()
{
    int marks;
    cin >> marks;

    if (marks >= 75)
        cout << "Grade A";
    else if (marks >= 60)
        cout << "Grade B";
    else if (marks >= 45)
        cout << "Grade C";
    else if (marks < 30)
        cout << "Grade D";
    else
        cout << "No grade specified";

    return 0;
}
```

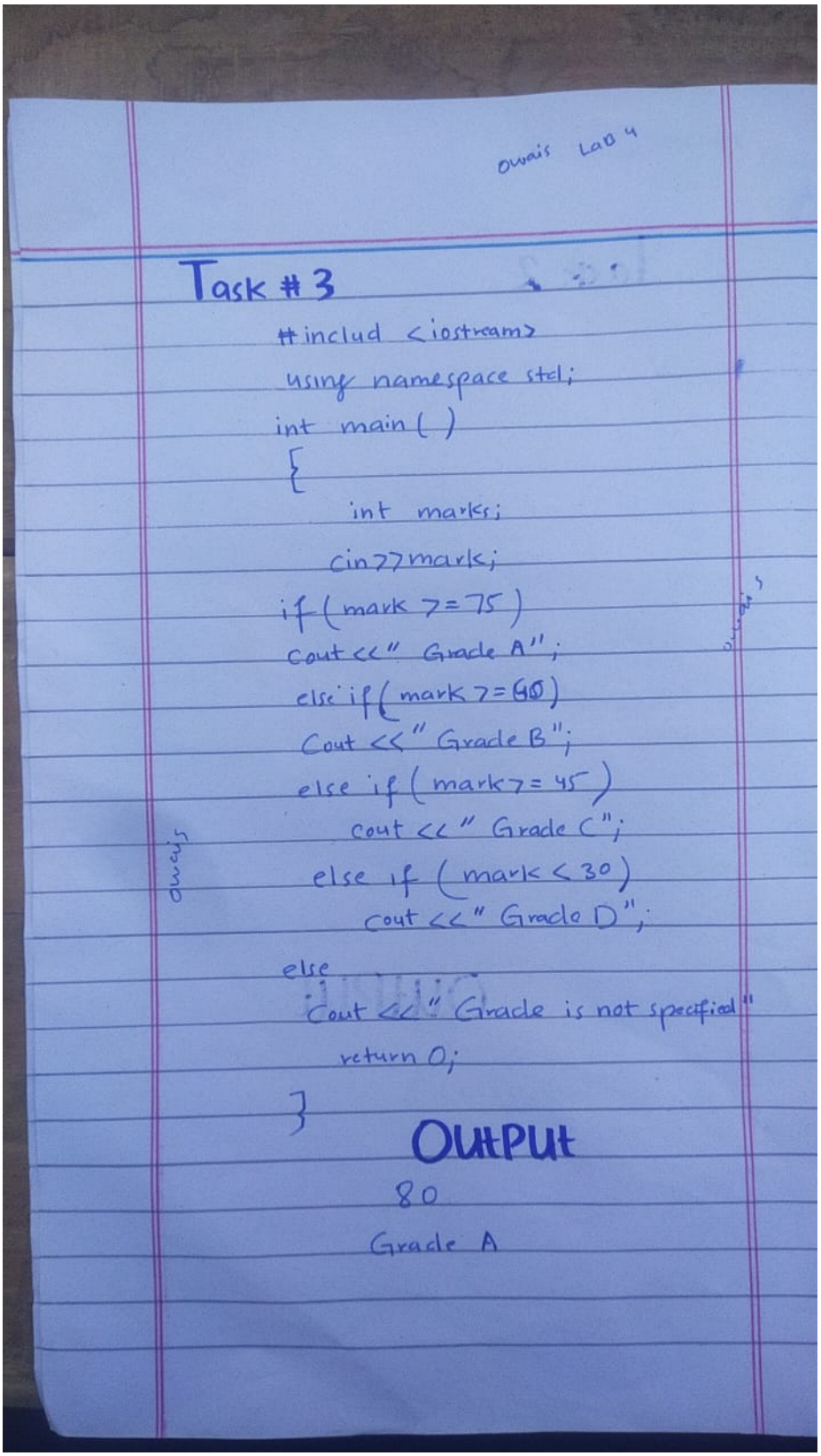
### Output

Focus folder in explorer (ctrl + click)

Terminal | Ports | Code

```
PS E:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles>c
d "e:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles\"
 ; if ($?) { g++ Task_3.cpp -o Task_3 } ; if ($?) { .\Task_3 }
80
Grade A
PS E:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles>
```

**Handwritten Notes**

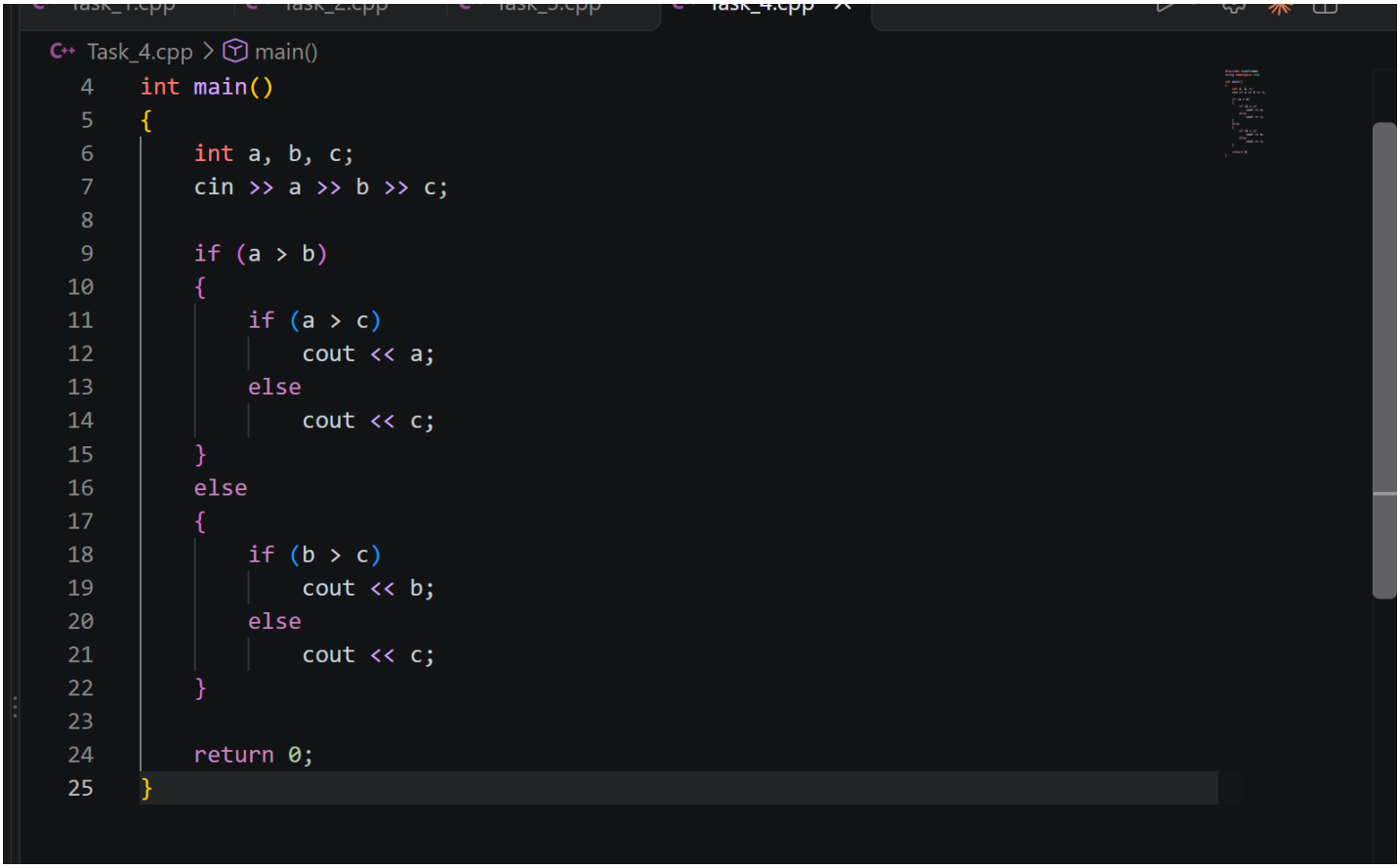


## Task 4:

Find Largest Number Using Nested if...else statement.

Check whether the number entered by the user is positive or not. If it is positive then calculate how many digits the number have.

### Input

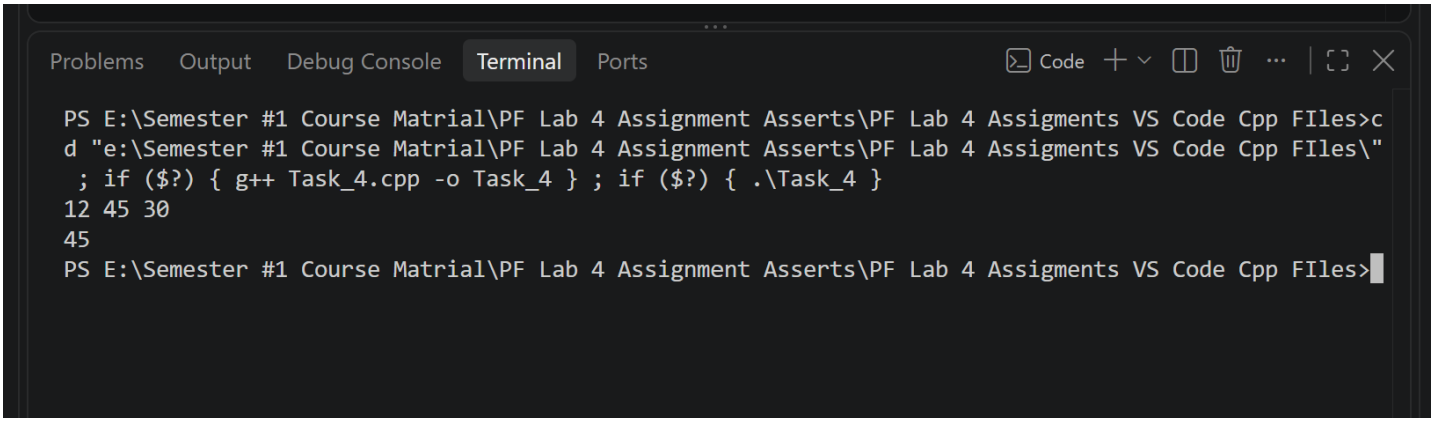


```cpp
int main()
{
    int a, b, c;
    cin >> a >> b >> c;

    if (a > b)
    {
        if (a > c)
            cout << a;
        else
            cout << c;
    }
    else
    {
        if (b > c)
            cout << b;
        else
            cout << c;
    }

    return 0;
}
```

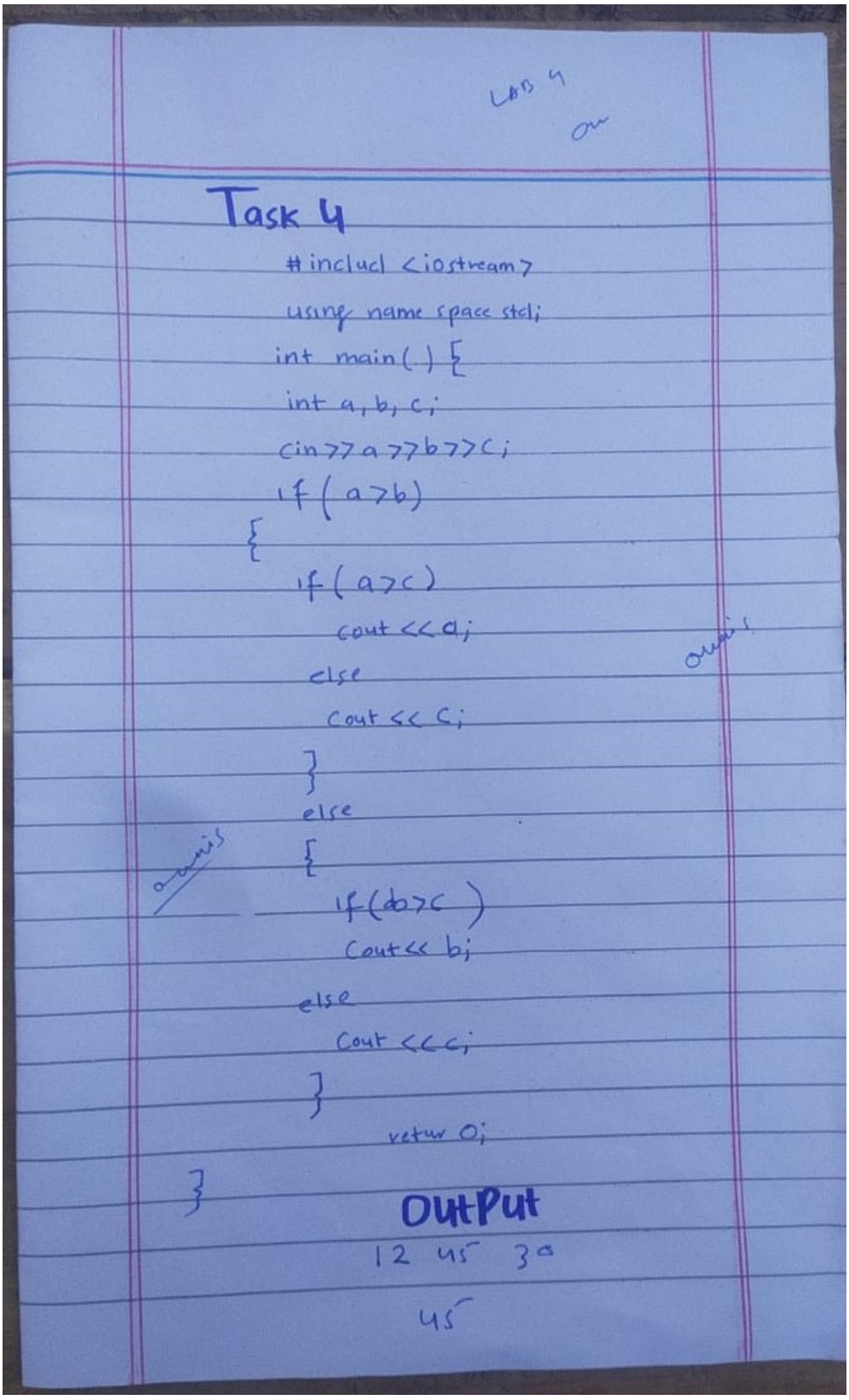
### Output



```
Problems  Output  Debug Console  Terminal  Ports

PS E:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles>c
d "e:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles\"
 ; if ($?) { g++ Task_4.cpp -o Task_4 } ; if ($?) { .\Task_4 }
12 45 30
45
PS E:\Semester #1 Course Matrial\PF Lab 4 Assignment Asserts\PF Lab 4 Assigments VS Code Cpp FIles>
```

**Handwritten Notes**



Lab 4
Owais

Task 4

```
#includ <iostream>
using name space std;
int main( ) {
int a, b, c;
cin >> a >> b >> c;
if ( a > b)
{
  if ( a > c)
    cout << a;
  else
    cout << c;
  }
  else
  {
    if (b > c )
      Cout << b;
  else
    Cout << c;
  }
    retur 0;
}
```

Owais

Owais

OutPut

12 45 30

45